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Palaeomagnetic dating of pseudotachylyte formation in the Lewisian complex

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Synopsis

Magnetic remanence is employed to date episodes of pseudotachylyte formation within the Lewisian terrane. In the Southern Zone on the mainland directions of magnetisation correlate with late stages of post-Laxfordian uplift and imply that these examples were either contemporaneous with the LM3 metamorphism or with early Torridonian rifting. In the Outer Hebrides the pseudotachylytes yield both mid-Proterozoic and Caledonian directions of magnetisation. The Proterozoic examples were formed late in the post-Laxfordian uplift cycle while the Palaeozoic examples are linked to formation of the Outer Hebrides thrust zone and presumably to the Caledonian thrusting on the mainland. The latter examples imply that differential uplift of the order of several kilometres has taken place between the Outer Hebrides and the Mainland since Caledonian times; their presence here explains why Torridonian and Lower Palaezoic cover rocks are not preserved in the former area.

Introduction

Pseudotachylytes, the flinty crush rocks generally attributed to frictional melting involving high strain rates during ancient seismic events, are found in two contrasting situations within the Lewisian metamorphic complex. Firstly, they are widely present in the Outer Hebrides south of Harris within, and on either side of, the Outer Hebrides thrust zones; they occur here as ramifying veins ranging from microscopic in size up to several metres in thickness (Jehu and Craig 1924; Francis and Sibson 1973). Secondly, they are widely distributed within the Southern Zone on the mainland (Park 1961) in WNW trending crush belts formed after the main Laxfordian metamorphic events and pegmatite emplacement in this area (*c.* 1900–1500 Ma), but prior to final uplift of the metamorphic complex and deposition of the overlying Torridonian sediments (*c.* 1000 Ma).

Whilst formation of the mainland pseudotachylytes can be constrained to the interval *c.* 1500–1000 Ma from field evidence, the age of the Hebridean examples can only be dated between late Laxfordian events (*c.* 1500 Ma) and deposition of the Stornoway Beds of probable Permo–Triassic age (Storetvedt and Steel 1977). The evidence for a more precise age assignment is conflicting: the geometrical similarity of the Outer Hebrides thrust zone with the Caledonian thrusts on the mainland, the occasional preservation of glass, and some K/Ar age dates, all

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suggest that pseudotachylyte formation here is to be linked to the Caledonian orogeny (Kursten 1957) and is presumably broadly contemporaneous with the Moine thrusting (i.e. 480–427 Ma, Mendum 1979). There are three difficulties with this interpretation:

(i) The Outer Hebrides thrust zones show some important contrasts with the Caledonian thrusts on the mainland (Francis and Sibson 1973); in particular they comprise a number of curved and interlinked planes, and the distributed development of pseudotachylyte (often not directly linked with the fault planes) contrasts with the absence of pseudotachylyte but the extensive development of mylonite within the Moine Thrust Belt.

(ii) The pseudotachylytes formed at appreciable depths in the crust (although at levels where deformation was still brittle) which were probably of the order of 6–10 km (Park 1961, Philpotts 1964, Francis and Sibson 1973); a Caledonian age would imply a substantial differential uplift has taken place between the Outer Hebrides and the mainland, where Phanerozoic platform successions are preserved near present sea level.

(iii) The mylonites of South Uist have been interpreted by Coward (1969) as phyllonites associated with the F3 Laxfordian deformation; this would imply that the pseudotachylytes (which are spatially associated here with the mylonites) are actually of late Laxfordian age.

Method

Owing to their fine grain size and variable alteration, pseudotachylytes are poor material for radiometric investigations. However, they generally possess a strong and stable remanent magnetism apparently because they contain abundant fine grains of ferromagnetic minerals in the single or pseudosingle domain size range ($<30\ \mu m$); these grains appear to have formed and acquired a magnetism in the ambient field over intervals of the order of seconds or minutes during which the pseudotachylytes chilled (Francis and Sibson 1973). Provided that this primary remanence can be recovered, the age of acquisition may be deduced by comparing the direction of magnetism with the predicted path of the palaeofield direction with respect to the sampling area between the time limits defined by geological evidence. This approach has been applied to the dating of pseudotachylyte formation in the Nagssugtoqidian mobile belt in East and West Greenland (Beckmann 1983, Piper 1981).

Samples for this study of Lewisian pseudotachylytes have been collected by field drilling at nine separate sites in veins in the vicinity of the Outer Hebrides thrust zone (Fig. 1) with each site comprising a number of separately oriented 2·4 cm diameter cores. A single site distributed over *c.* 100 metres was sampled at grid reference [NG 1835 8722] in the Southern Zone of the Lewisian on the mainland. Orientations were normally made with the samples still *in situ* by sights

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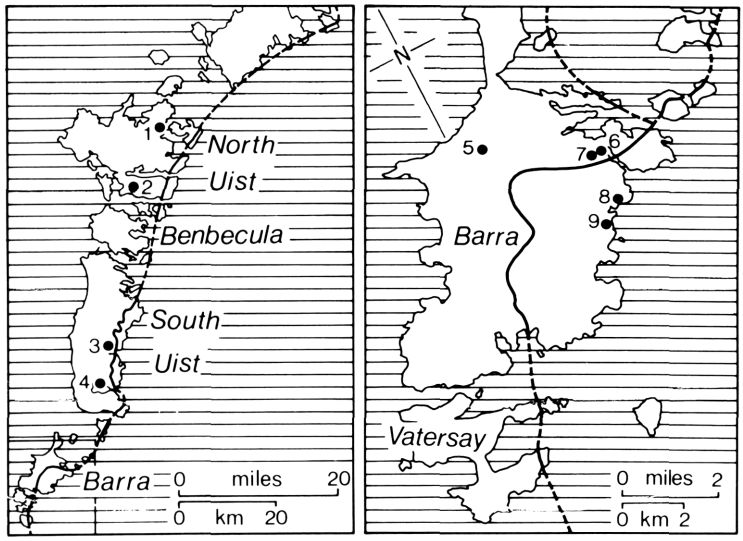


FIG. 1. Outline maps of the islands comprising the Uists and Barra in the Outer Hebrides showing palaeomagnetic sampling sites. The solid line defines the outcrop of the Outer Hebrides thrust zone after Francis and Sibson (1973).

on the Sun in conjunction with magnetic bearings and topographic sights. In the laboratory samples were progressively demagnetised either in alternating field (a.f.) steps of 5 or 10 milli Tesla (mT) to peak fields of 140 mT, or by progressive thermal treatment to peak temperatures incremented in intervals of 100°C to 500°C and then in reduced steps to the Curie points of the magnetic carriers. Initial experiences showed that the components of remanence in these rocks could be properly recovered as convergent vectors defined on orthogonal plots by a.f. demagnetisation, and the majority of the collection was accordingly treated ty this method.

RESULTS

Although this study is concerned with directions of magnetic remanence rather than magnetic properties, a noteworthy feature of the latter is the high Q_n value (ratio of remanent to induced magnetism) observed in this collection. Most values are greater than 10 and some are as high as 100. Such high mean values are often representative of fine grained magnetite and are most typically recorded in chilled igneous material such as pillow lavas (Ade Hall 1964; Watkins and Paster 1971). Magnetite carriers are confirmed by the response to a.f. cleaning and blocking temperature spectra below 580–600°C.

Seventeen cores drilled from the site in the Southern Zone on the mainland were stable to cleaning treatment and the majority illustrate single component

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behaviour with NW directions of positive inclination (e.g. Fig. 2a). In other cases a lower blocking temperature component is removed before the NW component is completely unblocked close to the Curie point of pure magnetite (Fig. 2b). A minority of samples yielded a more westerly direction (Fig. 2c); this direction was present as a lower blocking temperature component (<300°C) in some other cores which yielded the NW positive component in the higher blocking temperature

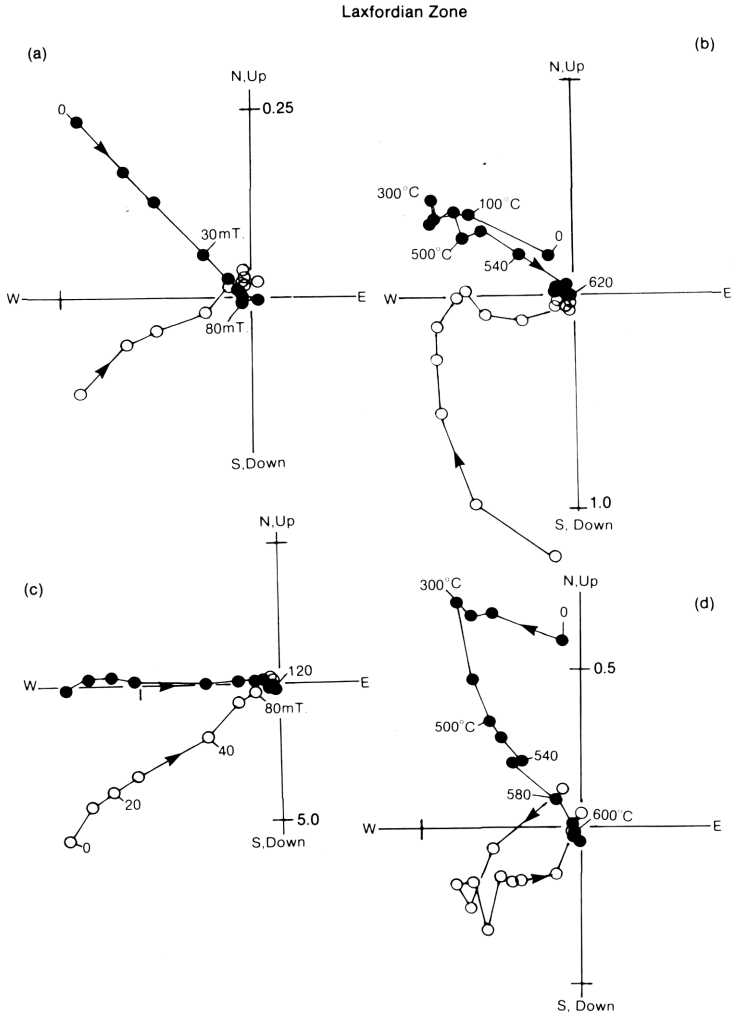


FIG. 2. Examples of a.f. and thermal demagnetisation behaviour of pseudotachylytes from the Southern Zone of the mainland illustrated as orthogonal projections. The projections of the remanence vector onto the horizontal plane are shown by the solid symbols and the projections onto the vertical plane are shown by the open symbols. The axes are calibrated in units of remanent intensity ($\times 10^{-5}$ A m^2 kg^{-1}).

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fraction (Fig. 2d). This component may therefore record both a younger overprint and a later episode of pseudotachylyte formation. Eleven samples falling in the NW positive (single polarity) group combine to yield a mean of $D = 313°$, $I = 28°$ ($\alpha_{95} = 8°$, Table 1). This direction lies within the sequence of directions representing the primary (A1 to A5) uplift magnetic remanences within the Southern Zone of the mainland Lewisian (Fig. 4, and Smith and Piper 1982).

The results from the Uists and Barra are more varied. Although most typically single component vectors are recovered, they often show more than one direction within the same sampling site. Some of these components are northerly or southerly directed with shallow inclinations (Fig. 3a and b); they are referred to

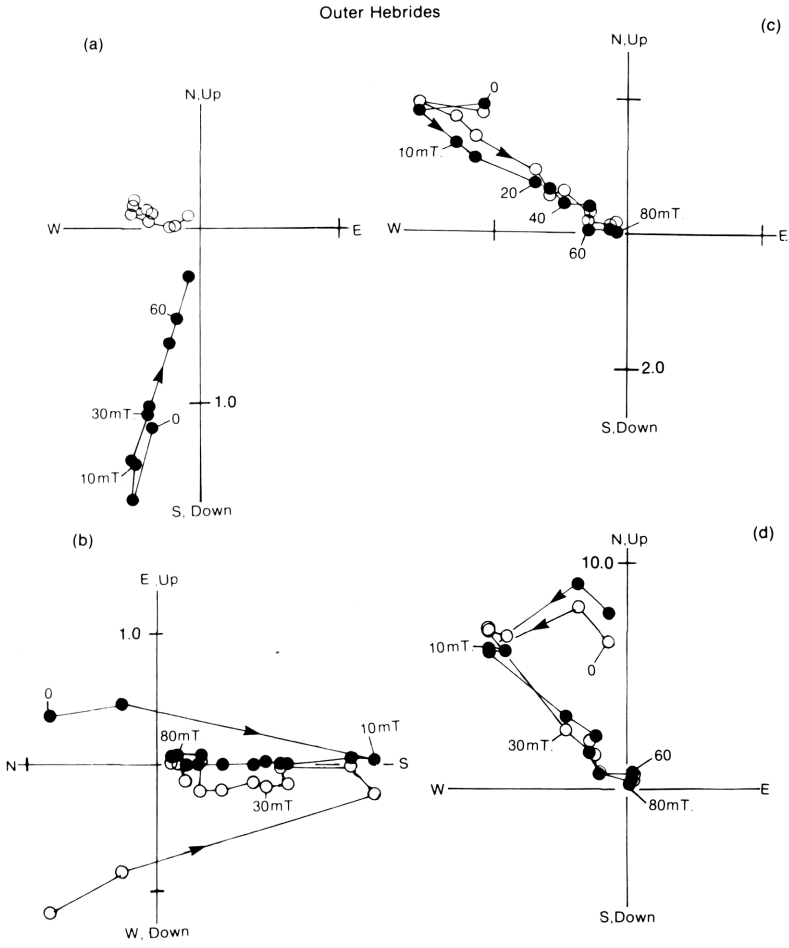


FIG. 3. Examples of a.f. demagnetisation behaviour of pseudotachylyte samples from the Outer Hebrides. Symbols are as for Fig. 2.

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here as B directions. Other samples yield NW declinations and inclinations which (unlike the components with similar declinations in the Southern Zone) are usually negative (Fig. 3c); these are referred to here as A directions. On the Uists, sites 3 and 4 yield components belonging to both groups, the remaining sites yield components belonging to group B with the majority being of reversed polarity (Table 1). On Barra three sites (5, 7, and 9) yield components belonging to group A and four sites yield components belonging to group B, with site 7 showing both components and two samples from site 9 yielding a B direction. In three samples the B direction is recovered following subtraction of an A component residing in a lower coercivity fraction (Fig. 3d).

Interpretation

The components comprising the majority population of directions from the Southern Zone have inclinations ranging from +12 to +50° and contrast with the NW negative/SW positive directions of magnetisation (T2 in Fig. 4) recovered from the overlying Torridonian sediments. They are not dissimilar from directions of magnetisation in the Stoer Group occurring further to the north (T1 in Fig. 4). We have employed palaeomagnetic conglomerate tests to demonstrate that significant magnetic overprinting of the Lewisian has not taken place in Lewisian amphibolites during Torridonian and later times (Poppleton and Piper 1988). This conclusion does not specifically apply to the pseudotachylytes, but in view of their high stability we suggest that their magnetite-held remanence is most likely to be post-Laxfordian in origin. In fact the mean direction plots on the sequence of post-Laxfordian components between the A3 and A4 mean directions from the Lewisian country rocks (Fig. 4). Poppleton and Piper (1988) correlate the A4 and A5 directions with the retrogressive LM3 metamorphism of Cresswell and Park (1973) dated at *c.* 1400 Ma by Moorbath and Park (1971) and the present surface may have reached levels compatible with pseudotachylyte formation in the post-Laxfordian uplift at approximately this time. A younger age coeval with deposition of the Stoer Group (cf. T1 in Fig. 4) is also possible and is indeed more probable because the greenschist facies metamorphism of LM3 does not appear to have affected the pseudotachylytes (Park 1961). A link with immediate post-Grenville events in the Laurentian Shield and the rifting leading to Torridonian sedimentation (Stewart 1975) is therefore implied (Smith, Stearn and Piper 1983; Piper 1987a).

The A group of directions from the Outer Hebrides (which, with the exception of one sample, is of single polarity) correlates only with the youngest part of the Proterozoic palaeofield migration path for NW Scotland defined by the post-Laxfordian uplift remanences (A5 in Fig. 4); it may also accord with the poorly-defined younger extension to this path indicated by some younger sporadic overprints in the Central Zone of the mainland (B in Fig. 4, Piper 1987b). A

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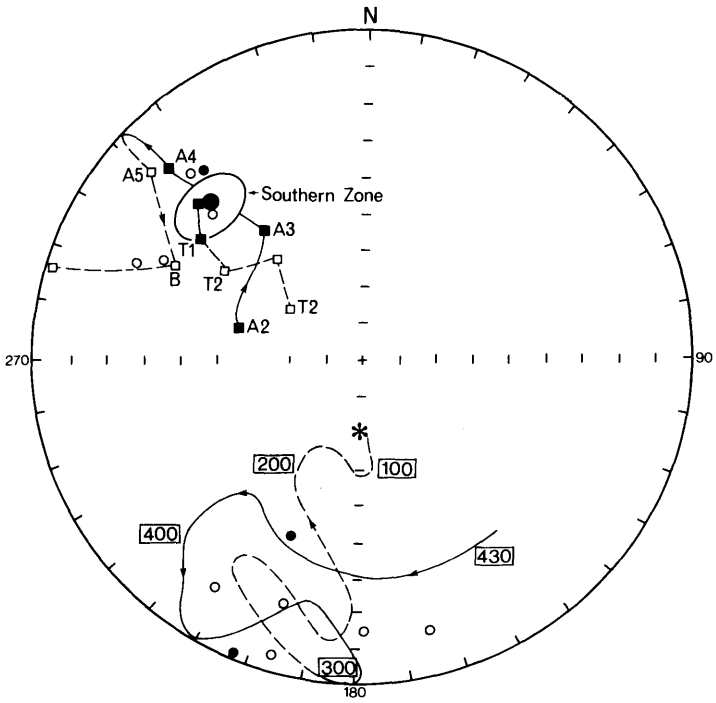


FIG. 4. Equal area projection showing directions of magnetisation from this study (circles) compared with predicted Proterozoic and Phanerozoic palaeofield directions (squares) for NW Scotland. Solid symbols and solid lines are lower hemisphere plots and open symbols and dashed lines are upper hemisphere plots. The directions A2 to A5 are representative directions defining the sequence of uplift magnetisations in the Lewisian terrane with the extension to B defined by sporadic overprints (after Smith and Piper 1982 and Piper 1987b). T1 and T2 are representative Lower and Upper Torridonian field directions respectively. The path commencing at 430 Ma is the Phanerozoic reversed field path predicted from other British and European data (Piper 1987a). Note that the direction of site 1 has been reversed in this figure.

post-Laxfordian and pre-Torridonian age is therefore also applicable to these examples although they cannot be contemporaneous with the examples in the Southern Zone. The population of B directions is not collectively well grouped and includes both polarities (Table 1). This suggests that the pseudotachylytes were formed by numerous events over a long interval of time involving both field reversals and apparent polar wander; this would be compatible with formation of the thicker examples by many small incremental slip episodes (Francis and Sibson 1973). The directions of magnetisation comprising the B group do not correlate with the established Proterozoic palaeofield migration path for NW Scotland (Fig. 4) but they plot instead close to the Palaeozoic path. The predicted direction of the reversed geomagnetic field direction for NW Scotland from 430 Ma is shown in Fig. 4 (after Piper 1987a) to illustrate this point. Unfortunately the pre-430 Ma

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TABLE 1

Palaeomagnetic results from pseudotachylyte veins in Lewisian terrane.

Location	N	D(°)	I(°)	R	Polarity
Southern Zone, Mainland	11	313	28	10·7	S
		(Palaeomagnetic pole: 233·4°E, 35·2°N dp/dm = 5·0/9·1)			
Uists, site 1	7	344	13	6·9	S
2	4	202	2	3·8	M
3(i)	3	201	39	2·8	M
3(ii)	3	315	−32	2·8	S
4(i)	2	189	3	1·9	S
4(ii)	6	319	22	5·6	S
Barra, site 5	5	292	−22	4·9	M
6	5	196	−18	4·9	S
7(i)	8	295	−29	7·5	S
7(ii)	4	212	−15	3·9	M
8A	3	178	−15	2·9	S
8B	8	196	−6	7·5	S
9	6	317	−20	5·5	S

D and I are the mean declination and inclination calculated from N samples, and R is the magnitude of the resultant vector. Sites subdivided as A and B are in the same vicinity but separated by more than 100 metres. S = single polarity and M = mixed polarity.

path spanning the interval of Caledonian times during which the mainland thrusts were developing is not yet defined from the Orthotectonic Caledonides, although data (480–440 Ma) from the Paratectonic Caledonides define directions not dissimilar from the 430–300 Ma path illustrated in this figure. A Caledonian age for these pseudotachylytes is therefore indicated by the palaeomagnetic evidence.

In view of the range of palaeomagnetic directions which has emerged from the pseudotachylytes on the Uists and Barra, we consider that a larger and more widely distributed sample is required to evaluate their full significance. However, it appears that pseudotachylytes of both mid-Proterozoic and Caledonian age are present here with the latter being prominent close to the Outer Hebrides thrust zone; this helps to explain the conflicting field evidence noted in a preceding section. A Caledonian age for the thrust zone requires that this area was at a crustal depth of the order of 5 km during these times. Appreciable post-Caledonian uplift is therefore implied and was presumably essentially complete by the time of deposition of the Stornoway Beds. This post-Caledonian uplift could explain why the Torridonian and Lower Palaeozoic cover sequences represented on the mainland are no longer preserved in the Outer Hebrides.

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